
Biographical Sketch

Personal Information

- **Full name:** Rajsekhar Mohapatra
- **Nationality:** Indian
- **Work address:** Department of Astrophysical Sciences, Peyton Hall, 4 Ivy Lane, Princeton University, Princeton, NJ 08544.
- **Email:** rmohapatra@princeton.edu
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Education

- **BSc in Physics**, Indian Institute of Science, Bengaluru, India, Thesis title: “*Studying turbulence in the intracluster medium*”, Supervisor: Prof. Prateek Sharma, **Aug 2014–June 2018**.
- **Time off** after undergraduate programme, before commencing PhD programme **July–Sep 2018**
- **PhD in astronomy**, Australian National University, ACT, Australia, Thesis title: “*The Role of Stratification, Turbulence and Cooling in the Intracluster Medium*”, Supervisor: Prof. Christoph Federrath, **Oct 2018–Sep 2022**.

Work Experience

- **Current position:** Postdoctoral research associate, Princeton University, Princeton, New Jersey, USA, **Sep 2022–present**.
Current research projects: Heating due to type Ia supernovae in massive ellipticals, Effect of Magnetic fields on Turbulent Radiative Mixing Layers, Multiphase turbulence in galactic halos.
- Software developer, AITC, Australian National University, Canberra, Australia, **June–Sep 2022**.
- Tutor in Physics, Australian National University, Canberra, Australia, **Jan 2020–May 2021**.

Recent Publications

- *The Type Ia Supernova and Asymptotic Giant Branch Stellar Ejecta-regulated Interstellar Medium of Massive Galaxies*, Mohapatra, R., Quataert, E, Fielding, D, and Guo, M, APJ, 2025, Vol 989, Issue 1, Page 103, doi:10.3847/1538-4357/adea6c.
- *Multiphase Gas in Elliptical Galaxies: The Role of Type Ia Supernovae*, Mohapatra, R, and Quataert, E, APJ, 2024, Vol 965, Issue 2, Page 105, doi: 10.3847/1538-4357/ad2940.
- *Multiphase condensation in cluster halos: interplay of cooling, buoyancy and mixing*, Mohapatra, R, Sharma, P, Federrath, C and Quataert, E, MNRAS, 2023, Vol 525, Issue 3, Pages 3831-3848, doi: 10.1093/mnras/stad2574.
- *Velocity structure functions in multiphase turbulence: interpreting kinematics of H α filaments in cool core clusters*, Mohapatra, R, Jetti, M, Sharma, P and Federrath, C; MNRAS, 2022, Vol 510, Issue 2, pp.2327-2343, doi: 10.1093/mnras/stab3429.
- *Turbulent density and pressure fluctuations in the stratified intracluster medium*, Mohapatra, R, Federrath, C and Sharma, P, MNRAS, 2021, Vol 500, Issue 4, Pages 5072-5087, doi: 10.1093/mnras/staa3564.

Research Interests and Expertise

My research focuses on the physics of multiphase gas in galaxy clusters and massive galaxies, with particular emphasis on turbulence, thermal instability, and feedback processes such as Type Ia supernovae and AGN heating. My work combines high-resolution magnetohydrodynamic simulations with theoretical modeling to understand the formation and evolution of cold gas in the intracluster medium. I have made significant contributions to the study of turbulent mixing, stratified turbulence, and the kinematics of filamentary structures in the intracluster medium.

Synergistic Activities

- **Conference organization:** Member of organizing committee for PCTS workshop “Synergistic approaches to particle transport in magnetized turbulence”, 2024.
- **Mentoring:** Supervised 3 high school students, 3 summer internships, 1 Bachelors thesis, 1 Masters thesis, 2021–present.
- **Referee:** for journals A&A, APJ, Astronomy and Computing, MNRAS, Frontiers, OJAP.
- **Committee membership:** Member of Climate committee, Department of Astrophysical Sciences, Princeton University, 2024–present.
- **Public outreach:** Member of public observing team at Princeton University, 2023–present, former outreach assistant at Mount Stromlo Observatory, Australia, 2019–2022.

Collaborators

Alankar Dutta, Andrew Fabian, Ang Liu, Benjamin D Wibking, Brian W O’Shea, Chang-Goo Kim, Christoph Federrath, Ciro Pinto, Deovrat Prasad, Dominique Eckert, Drummond Fielding, Efrain Gatuza, Eliot Quataert, Forrest W Glines, Greg Bryan, Helen Russell, Jeremy Sanders, John Zuhone, Lachlan Lancaster, Marcus Bruggen, Martin Fournier, Minghao Guo, Mrinal Jetti, Philipp Grete, Prateek Sharma, Sanskriti Das, Stephen Walker, Tejahni Desire.

Recent conferences and programmes attended

- *Modeling of Multiphase Astrophysical Media workshop*, Schloss Ringberg, Germany, Nov 2025.
- *National Astronomy Meeting*, Durham, UK, July 2025.
- *Baryons Beyond Galactic Boundaries*, IUCAA, Pune, India, Dec 2024.
- *Toward a Holistic Understanding of the Multi-scale, Multiphase Circumgalactic Medium*, Aspen Center for Physics, Colorado, USA, Aug 2024.
- *ICM Theory and Computation workshop*, University of Michigan, USA, June 2024.
- *Turbulence in the Universe*, KITP, Santa Barbara, USA, Feb 2024.

Successful computing proposals

- 2023: 3 million ACCESS credits (45000 GPU hours) through the ACCESS scheme as PI.
- 2024: 13 million ACCESS credits as co-PI.
- 2025: 20 million ACCESS credits as co-PI; 800,000 Node hours on FRONTIER as co-I.

Coding and supercomputing skills

- **Programming Languages:** C, C++, FORTRAN, Python, L^AT_EX, VisIt, Shell scripting.
- **Magnetohydrodynamics code:** AthenaX, FLASH, PLUTO.
- **Supercomputers used:** Frontier (ORNL, USA); Della, Stellar (Princeton, USA); Delta, DeltaAI (Illinois, USA); Lumi (Finland); Gadi, Raijin, (Australia); Sahashrat (India).